

The long-term effect of indicating AI fallibility on automation bias in mental health application users.

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Figure 1: Seattle Mariners at Spring Training, 2010.

Abstract

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CCS Concepts

• **Information systems** → Users and interactive retrieval; Evaluation of retrieval results; • **General and reference** → Surveys and overviews; • **Computing methodologies** → Knowledge representation and reasoning; **Artificial intelligence**; • **Social and professional topics** → Age; Gender; • **Human-centered computing** → Human computer interaction (HCI); **Empirical studies in interaction design**; *Visualization*; **Empirical studies in HCI**.

Keywords

Mental Health, AI, Artificial Intelligence, FWS, STUD, Automation Bias, Fallibility, Mental Health Application, AI Assistant, Long-Term, Long-Term Effects, Priming, Trust, Confidence

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1 Introduction

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The LaTeX typesetting markup language is specially suitable for documents that include mathematics.

2 Definitions

2.1 Participants

We define the following sets:

Groups $G := \{\text{Active, Control}\}$,

Case Vignettes $C := \{\text{Amina, Aylin, Lena, Martina, Mathias}\}$,

Truthfulness $T := \{\text{Truth, Untruth}\}$.

A *round* is a pair consisting of a case vignette and a truthfulness label. Hence, the set of all rounds is

$$R := C \times T.$$

A *session* consists of exactly three rounds. Formally, a session is an ordered triple

$$s = (r_1, r_2, r_3) \in R^3.$$

However, not every element of R^3 is considered valid. A valid session must satisfy the following constraints:

$$|\{i \in \{1, 2, 3\} \mid r_i = (c, \text{Truth}) \text{ for some } c \in C\}| = 1,$$

i.e. exactly one round in the session is labelled Truth.

Furthermore,

$$\forall i, j \in \{1, 2, 3\} : i \neq j \Rightarrow c_i \neq c_j,$$

where $r_i = (c_i, t_i)$. That is, no case vignette may occur more than once within the same session.

We denote the set of all valid sessions by

$$S \subseteq R^3.$$

A participant is a triple

$$p = (g, s_1, s_2) \in G \times S \times S,$$

where

$$g \in G, \quad s_1, s_2 \in S.$$

The two sessions of a participant must share exactly one case vignette. Formally,

$$|\{c \in C \mid \exists t_1, t_2 \in T : (c, t_1) \in \{r_1, r_2, r_3\} \wedge (c, t_2) \in \{r'_1, r'_2, r'_3\}\}| = 1,$$

where

$$s_1 = (r_1, r_2, r_3), \quad s_2 = (r'_1, r'_2, r'_3).$$

An example of a valid participant is

$$p = (\text{Active}, \\ ((\text{Amina, Truth}), (\text{Lena, Untruth}), (\text{Mathias, Untruth})), \\ ((\text{Martina, Truth}), (\text{Aylin, Untruth}), (\text{Mathias, Untruth}))).$$

2.2 Participant Allocation

To generate participants satisfying the constraints defined in the previous subsection, we employ a deterministic pseudo-random allocation algorithm, denoted by A_s (Participant Allocator).

For a fixed pseudo-random seed s , the algorithm defines a mapping

$$A_s : \mathbb{N}_0 \rightarrow P,$$

where

$$A_s(n) = p_n,$$

with

- P denoting the set of valid participants defined previously,
- $n \in \mathbb{N}_0$ denoting the participant index,
- $p_n \in P$ denoting the generated participant corresponding to index n ,
- $s \in \mathbb{N}$ denoting the global pseudo-random seed.

In the present study, the seed value $s = 403$ was used. The algorithm is deterministic with respect to the seed s , thereby ensuring reproducibility of the participant allocation.

The reference Python implementation is provided in the attached file `allocate.py`.

2.3 Automation bias

Automation bias is modelled as a continuous function that maps evaluation parameters to a scalar value in the interval $[0, 1]$, representing the degree of evidence that automation bias has occurred.

The function is defined as:

$$B : \{0, 1\} \times \{1, 2, 3, 4, 5\} \times [0, 100] \rightarrow [0, 1],$$

$$B(i, r, c),$$

where:

- $B(i, r, c) \in [0, 1]$ denotes the automation bias score,
- $i \in \{0, 1\}$ indicates correctness of the automated answer (1 = correct, 0 = incorrect),
- $r \in \{1, 2, 3, 4, 5\}$ denotes the evaluator's rating of the answer (1 = very bad, 5 = very good),
- $c \in [0, 100]$ denotes the evaluator's confidence in the rating.

2.3.1 Definition of the Bias Function. The function is defined as:

$$B(i, r, c) = (1 - i) \cdot \frac{r - 1}{5 - 1} \cdot \frac{c}{100}.$$

2.3.2 Interpretation. The term $(1 - i)$ ensures that automation bias is only considered when the automated answer is incorrect. If the answer is correct ($i = 1$), then $B(i, r, c) = 0$ regardless of the other parameters.

The term $\frac{r-1}{4}$ maps the rating to a normalized interval $[0, 1]$, where higher ratings indicate stronger perceived quality of the answer. In the context of automation bias, higher ratings for incorrect answers indicate a stronger tendency toward bias.

The term $\frac{c}{100}$ represents the evaluator's confidence in their rating. Higher confidence strengthens the evidence of automation bias.

By construction, $B(i, r, c) \in [0, 1]$.

2.3.3 Threshold for Minimal Automation Bias. We define a reference threshold based on a minimally biased configuration:

$$(i, r, c) = (0, 3, 50).$$

Substituting into the function yields:

$$\begin{aligned} B(0, 3, 50) &= (1 - 0) \cdot \frac{3 - 1}{4} \cdot \frac{50}{100} \\ &= 1 \cdot \frac{2}{4} \cdot 0.5 \\ &= 1 \cdot 0.5 \cdot 0.5 \\ &= 0.25. \end{aligned}$$

We therefore define the minimal automation bias threshold as:

$$B_{\text{threshold}} = 0.25, \quad \text{and we consider } B(i, r, c) > 0.25$$

as indicating evidence of automation bias.

3 Testing Methods

3.1 General

Our tests followed this formula:

- (1) Participant gives consent
- (2) Participant is given a short context description
"Wir haben einen App für eine KI für mentale Gesundheit vorbereitet. Sie kann Fälle von Patienten einschätzen und Vorschläge für Behandlungen geben."
- (3) Participant gives themselves a nickname
- (4) Participant answers demographics questions
 - Gender
 - Male
 - Female
 - Nonbinary
 - Other
 - Prefer not to say
 - Age
- (5) Question About AI
 - How do you view the use of Artificial Intelligence in terms of usefulness?
 Likert Scale $\{1, 2, 3, 4, 5\} \in \text{Not useful at all} \leftrightarrow \text{very useful}$
 - How do you view the use of Artificial Intelligence in terms of acceptability?
 Likert Scale $\{1, 2, 3, 4, 5\} \in \text{Very unacceptable} \leftrightarrow \text{very acceptable}$
 - How do you view the use of Artificial Intelligence in the field of mental health in terms of usefulness?
 Likert Scale $\{1, 2, 3, 4, 5\} \in \text{Not useful at all} \leftrightarrow \text{very useful}$

- How do you view the use of Artificial Intelligence in the field of mental health in terms of acceptability?
 Likert Scale $\{1, 2, 3, 4, 5\} \in \text{Very unacceptable} \leftrightarrow \text{very acceptable}$
- How frequently do you use Artificial Intelligence in the field of mental health?
 Likert Scale $\{1, 2, 3, 4, 5\} \in \text{Not at all} \leftrightarrow \text{very often}$
- How frequently do you use Artificial Intelligence otherwise?
 Likert Scale $\{1, 2, 3, 4, 5\} \in \text{Not at all} \leftrightarrow \text{very often}$

- (6) Participant is handed Case Vignette & Instructor sets up App and enters values into database
- (7) participant reads case vignette
- (8) participant signals to continue
- (9) participant gets App
- (10) (If Active) participant gets Disclaimer
- (11) (If Active) participant waits out and reads Disclaimer
- (12) participant sends Prompt
- (13) (If Active) participant gets inline Disclaimer
- (14) participant reads Answer
- (15) participant answers questions
 - How do rate the AI Answer?
 Likert Scale $\{1, 2, 3, 4, 5\} \in \text{very bad} \leftrightarrow \text{very good}$
 - How certain are you in your choice?
 $[0; 100]$
 - questions from UEQ+
 - Intuitive Use
 - Trustworthiness of Content
 - Response behavior
 - Response quality
 - Clarity
 - Risk Handling
- (16) repeat (6) 2 more times
- (17) end of study Session
- (18) 1 week passes
- (19) repeat (1)
- (20) remind of context
- (21) patient remembers nickname or is told if they forgot
- (22) repeat (5) to (16) but always without disclaimers

3.2 Session 1

3.3 Session 2

4 Results

Acknowledgments

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We also thank all testers who participated in the study and made this work possible.

LIST OF FIGURES

- 1 Seattle Mariners at Spring Training, 2010.

LIST OF TABLES

References

- [1] [SW] The CGAL Project, The Computational Geometry Algorithms Library (Coord.by CGAL Editorial Board), 1996. URL: <https://cgal.org/>.

A Palloc Python Implementation

[allocate.py](#)